

Date 20/01/22/20
 Class BSc 2nd year
 Roll No 68
 Shift Day

Physics Practical Sheet

Prithvi Narayan Campus

Experiment No 65
 Group P2
 Sub Physres
 Set L

Object of the Experiment (Block Letters) : TO CONSTRUCT AND VERIFY LOGIC GATES USING IC (INTEGRATED CIRCUIT).

APPARATUS REQUIRED:

- | | | |
|-------------------|----------------|------------------|
| (i) Multimeter | (ii) Wires | (iii) Breadboard |
| (iv) Power supply | (v) IC 7408 | (vi) IC 7432 |
| (vii) IC 7400 | (viii) IC 7402 | (ix) IC 7404 |

THEORY:

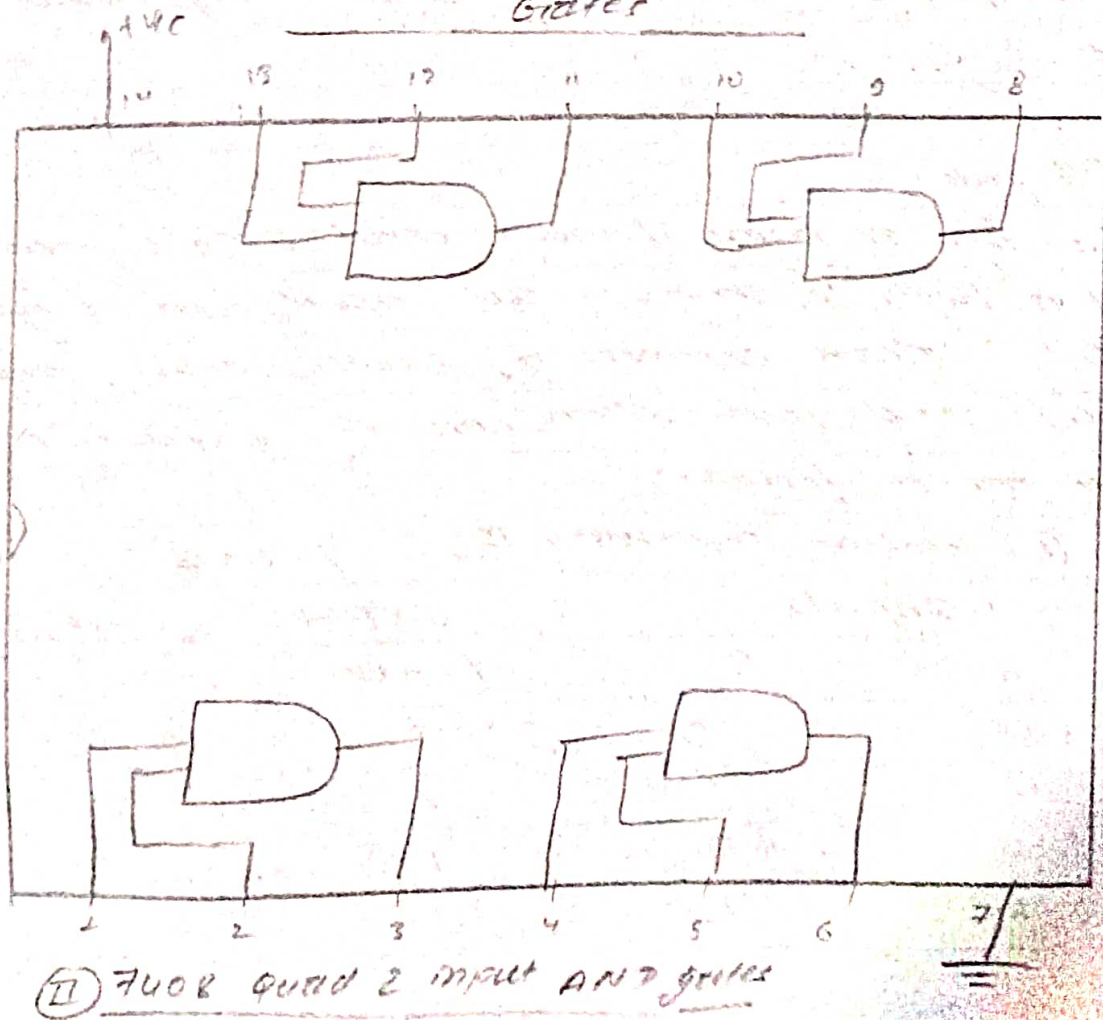
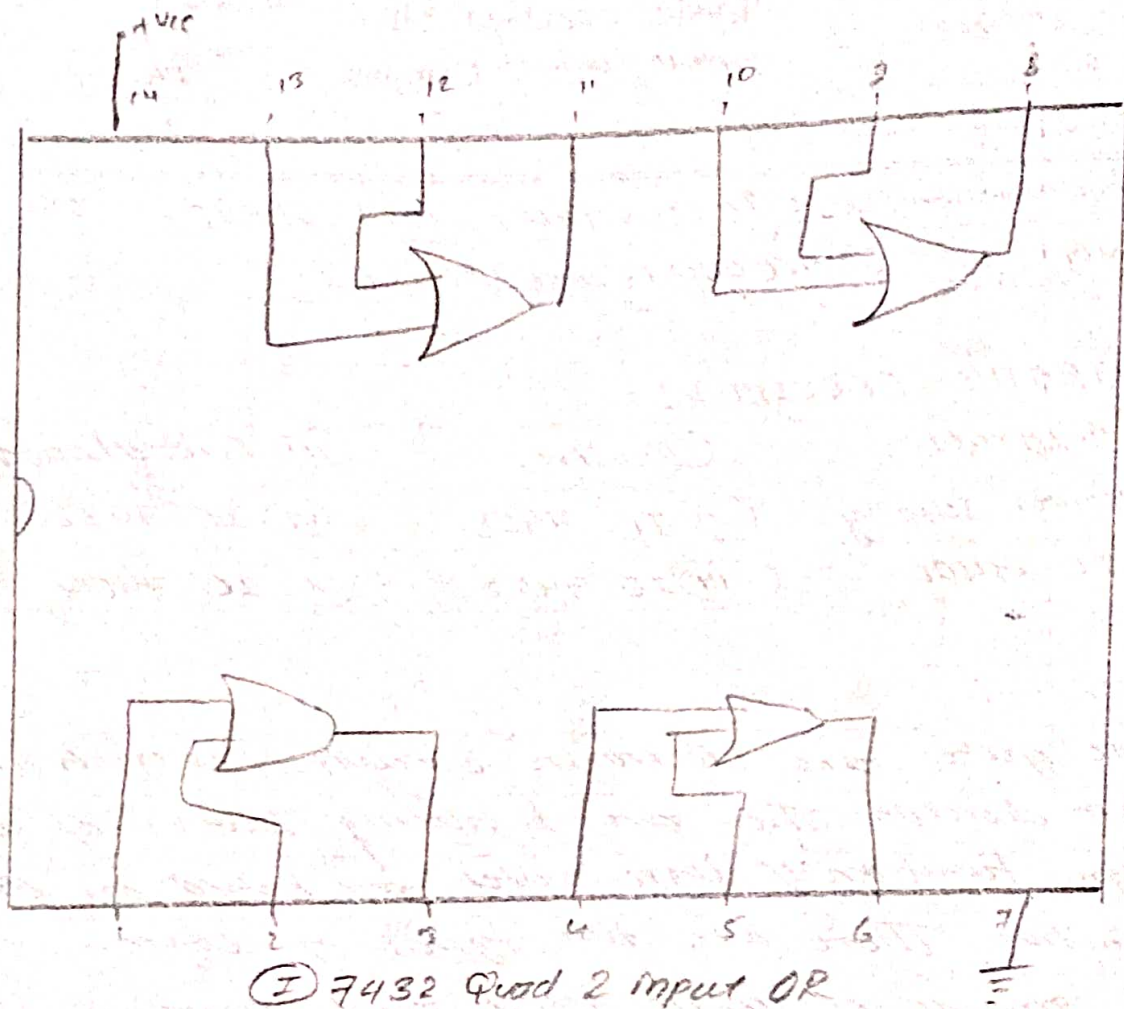
Logic gates are electronic circuits which makes logic decision. They are building blocks of digital system. Function of logic gates are based on Boolean equation. There are two types of logic-gate. They are simple or basic gates and compound logic gate.

(i) OR gate:

It has two or more inputs and single output. Its input is high if any or all input are high. Logic symbol, Boolean expression, truth table, practical circuit and realization circuit of two input OR gates are as follows:

The Boolean expression is, $Y = A + B$

S.N	Inputs		Output	Truth Table
	A	B	$Y = A + B$	
1.	0	0	0	
2.	0	1	1	
3.	1	0	1	
4.	1	1	1	



(II) AND gate:

It has two or more inputs and a single output.

Its output is high if all inputs are high. Logic symbol, Boolean equation, truth table, practical circuit diagram of AND gates are as follows:

The Boolean expression is, $Y = A \cdot B$

Truth Table

SN	Inputs		Output
	A	B	$Y = A \cdot B$
1.	0	0	0
2.	0	1	0
3.	1	0	0
4.	1	1	1

(III) NOT gate:

It has one input and one output. It reverse the nature of input. It converts the input to false and vice-versa.

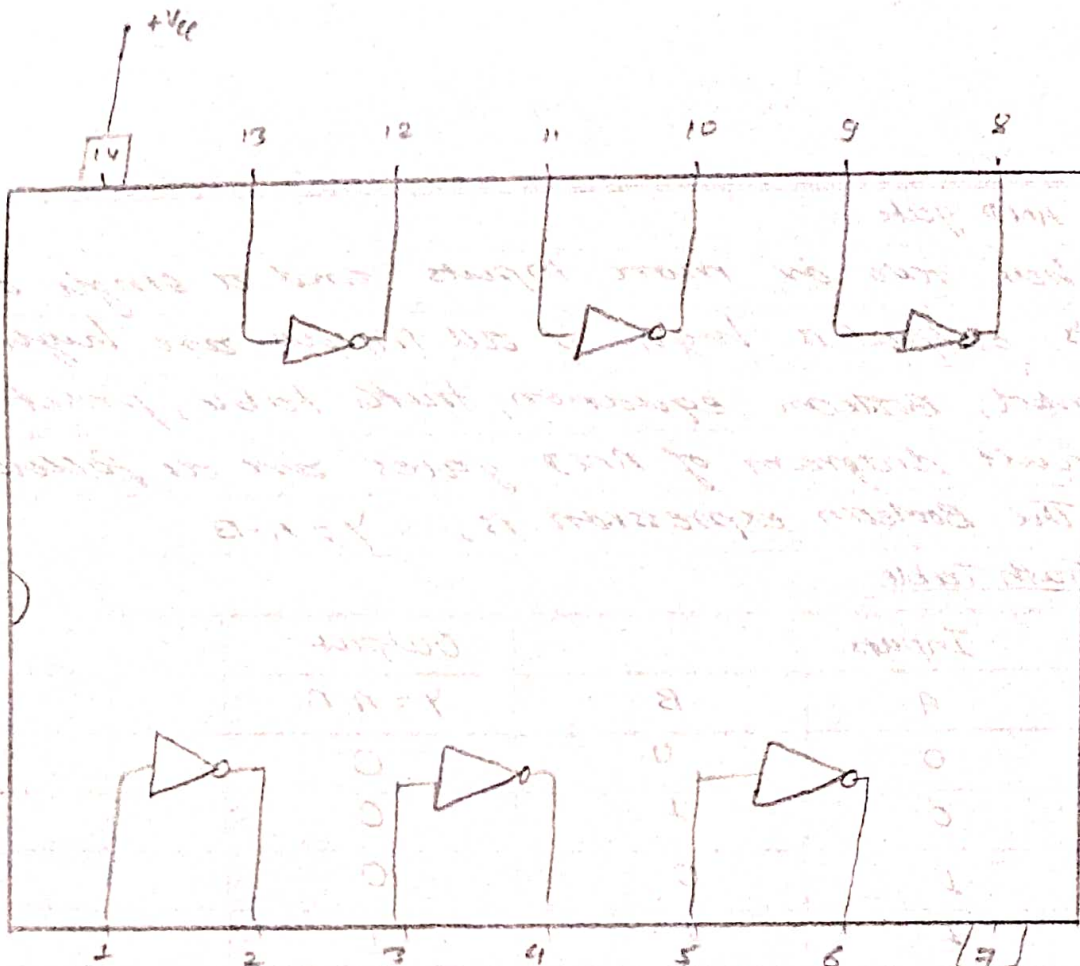
The Boolean expression is $Y = \bar{A}$.

Truth Table

SN	Input	Output
	A	$Y = \bar{A}$
1.	0	1
2.	1	0

(IV) NAND gate:

NAND gate is a type of gate that is formed by the combination of a AND and NOT gate i.e., if AND gate output is given to a NOT gate, the



III 7404 Hex NOT Gates



IV 7400 Quad 2 Input NAND Gates

combination forms NAND gate. The output due to the combination is equal to a output given by the combination of NOT, AND gates. Hence, the combination of these two gives a new gate.

The Boolean expression is, $Y = \overline{A \cdot B}$

Truth Table

S.N	Inputs		Output
	A	B	$Y = \overline{A \cdot B}$
1.	0	0	1
2.	0	1	1
3.	1	0	1
4.	1	1	0

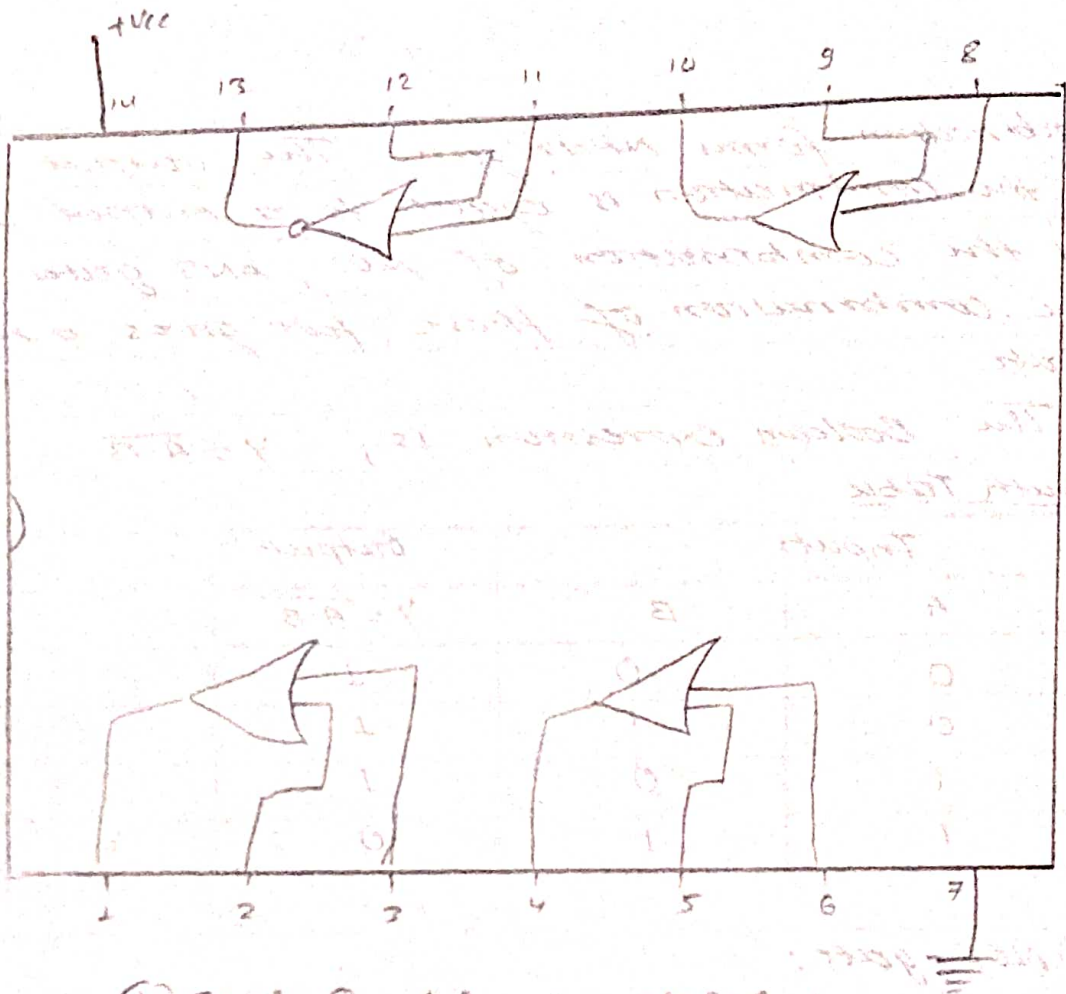
(✓) NOR gate:

NOR gate is the combination of NOT and OR gate. It behaves according to the truth table. A high output results if both the inputs to the gate are low and if one or both input is high, a low output result. NOR is the result of the negation of the OR operator.

The Boolean expression is, $Y = \overline{A + B}$

Truth Table

S.N	Inputs		Output
	A	B	$Y = \overline{A + B}$
1.	0	0	1
2.	0	1	0
3.	1	0	0
4.	1	1	0



④ 7402 Quad 2 input NOR Gates

OBSERVATION:

(I) OR gate (IC-7432)

S.N	Inputs		Output
	A	B	$Y = A + B$
1.	0	0	0
2.	0	4.74	4.72
3.	4.74	0	4.81
4.	4.74	4.74	4.74

(II) AND gate (IC-7408)

S.N	Inputs		Output
	A	B	$Y = A \cdot B$
1.	0	0	0.31
2.	0	4.19	0.30
3.	4.19	0	0.28
4.	4.19	4.19	3.50

(III) NOT gate (IC-7404)

S.N	Input	Output
	A	$Y = \bar{A}$
1.	0	4.05
2.	4.23	0

(IV) NAND gate (IC-7400)

S.N	Inputs		Output
	A	B	$Y = \overline{A \cdot B}$
1.	0	0	4.56
2.	0	4.60	4.57
3.	4.60	0	4.61
4.	4.60	4.60	0.01

(v) NOR gate (IC-7402)

S.N	Inputs		Output
	A	B	$Y = \overline{A+B}$
1.	0	0	4.13
2.	0	4.47	0.04
3.	4.47	0	0
4.	4.47	4.47	0

RESULT:

The basic and compound logic gates are verified using integrated circuit (IC).

PRECAUTION:

- (i) Wires should be connected properly.
- (ii) The apparatus should be connected carefully.
- (iii) Voltage should be noted correctly.